Supplementary Material

PhysioRisk: an interpretable survival-modeling framework for separating demographic and physiological mortality risk

Biogerontology

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Numerical values are transcribed from the cited canonical artifacts and rounded only for display; source artifacts retain full precision.

Supplementary Methods

Laboratory measurements were harmonized across NHANES cycles to prespecified units before residualization or score construction. PhysioRisk used creatinine in mg/dL. For PhenoAge NCP, albumin was expressed in g/L, creatinine in μmol/L, and glucose in mmol/L before application of the published coefficients.

PhenoAge NCP was calculated as:

−19.907 − 0.0336 × albumin (g/L) + 0.0095 × creatinine (μmol/L) + 0.1953 × glucose (mmol/L) − 0.0120 × lymphocyte percentage + 0.0268 × mean corpuscular volume (fL) + 0.3306 × red cell distribution width (%) + 0.00188 × alkaline phosphatase (U/L) + 0.0554 × white blood cell count (10^3/μL) + 0.0804 × age (years).

C-reactive protein was omitted. Albumin was converted from g/dL to g/L, creatinine from mg/dL to μmol/L, and glucose was harmonized across survey cycles to mmol/L before applying the coefficients. Glucose entered the predictor linearly.

Glucose was harmonized to mmol/L before calculation of PhenoAge NCP. For 2001–2002 and 2003–2004, the NHANES SI variable LBDSGLSI was used directly in mmol/L. For 1999–2000 and 2005–2016, LBXGLU was supplied in mg/dL and converted to mmol/L by multiplication by 0.05551.

Supplementary Table S1. Paired C-index contrasts in the temporal-validation cohort

Contrast	Cohort	ΔC (A − B)	95% CI low	95% CI high	Bootstrap proportion ΔC > 0	Replicates	Seed
CalibratedTotalRisk − DemoRisk	temporal_test	0.01762	0.01197	0.02366	1.0000	2000	20260821
CalibratedTotalRisk − PhenoAge NCP	temporal_test	−0.00026	−0.00729	0.00626	0.4775	2000	20260821
PhysioRisk − PhenoAge NCP acceleration	temporal_test	0.00884	−0.01160	0.02892	0.8205	2000	20260821

Note: Paired percentile intervals were obtained from 2,000 participant bootstrap replicates in the temporal-validation cohort. Contrast estimates and interval limits are shown to the manuscript's five-decimal reporting precision; the canonical source artifact retains full precision.

Provenance: outputs/physiorisk_v2_validation/paired_cindex_bootstrap_v2.csv.

Supplementary Table S2. Proportional-hazards diagnostics

Model	Cohort	N	Deaths	Time transform	Test statistic	P value	Material violation	Flag rule	Cox warning	Scaled-residual region summary
DemoRisk	temporal_test	6,739	416	rank	2.925	0.08720586	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=-0.087; middle_37_72m: mean=0.044; late_73m_plus: mean=0.099
PhysioRisk	temporal_test	6,739	416	rank	0.024	0.87657403	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=0.010; middle_37_72m: mean=-0.028; late_73m_plus: mean=0.042
CalibratedTotalRisk	temporal_test	6,739	416	rank	11.569	0.00067047	True	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=-0.145; middle_37_72m: mean=0.079; late_73m_plus: mean=0.154
PhenoAge NCP	temporal_test	6,739	416	rank	0.078	0.77993945	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=0.012; middle_37_72m: mean=-0.050; late_73m_plus: mean=0.086
PhenoAge NCP acceleration	temporal_test	6,739	416	rank	1.007	0.31566696	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=0.029; middle_37_72m: mean=-0.043; late_73m_plus: mean=0.031

Note: The prespecified descriptive flag was p<0.01 without multiplicity adjustment. The regional scaled-residual summaries are reported where present in the canonical artifact.

Provenance: outputs/physiorisk_v2_validation/ph_diagnostics_v2.csv.

Supplementary Table S3. Diagnostic log-time interaction analyses

Model	N	Deaths	Score scale	Time interaction	Reference time (months)	Log HR/SD at 60m	HR/SD at 12m	HR/SD at 60m	HR/SD at 96m	Interaction coefficient	95% CI low	95% CI high	P value	Direction
CalibratedTotalRisk	6,739	416	actual temporal-cohort SD	score_z * log(time_months/60)	60	1.286	2.935	3.617	3.844	0.12970	0.02929	0.23011	0.01135	effect_strengthens_over_time

Note: These analyses are diagnostic only and did not replace or refit the fixed predictors used in temporal validation. The interaction did not meet the prespecified p<0.01 descriptive flag (No).

Provenance: outputs/physiorisk_v2_validation/time_interactions_v2.csv.

Supplementary Table S4. Fixed-horizon calibration of CalibratedTotalRisk

Horizon (months)	Training baseline survival	Mean predicted mortality	KM observed mortality	Observed − predicted	O/E ratio	Global Cox slope	Global slope CI low	Global slope CI high	Horizon intercept	Horizon slope	IPCW Brier score	IPCW usable N	No temporal recalibration
60	0.983996	0.04845	0.04653	−0.00192	0.960	0.804	0.755	0.852	−0.267	0.895	0.0377	4,709	True
96	0.967042	0.08934	0.08826	−0.00108	0.988	0.804	0.755	0.852	−0.060	0.963	0.0604	1,518	True

Note: Absolute risks used the development-derived coefficients and baseline survival. Horizon-specific logistic calibration and Brier scores used temporal-test censoring weights for evaluation only; no temporal recalibration occurred.

Provenance: outputs/physiorisk_v2_validation/fixed_horizon_calibration_v2.csv.

Supplementary Table S5. Censoring-robust discrimination diagnostics

Score	Uno C, 5 years	Uno C, 8 years	AUC, 5 years	AUC, 8 years
DemoRisk	0.823	0.827	0.834	0.853
PhysioRisk	0.666	0.663	0.665	0.682
CalibratedTotalRisk	0.841	0.844	0.851	0.870
PhenoAge NCP	0.841	0.845	0.853	0.875
PhenoAge NCP acceleration	0.663	0.654	0.671	0.690

Note: The development cohort estimated the censoring distribution. These were diagnostics of fixed temporal-test predictions, not model-training steps.
Provenance: outputs/physiorisk_v2_validation/censoring_robust_discrimination_v2.csv.

Supplementary Table S6. Cycle-specific preselection biomarker missingness and cohort provenance

Cycle start year	Biomarker	Mortality-eligible N	Missing N	Missing (%)	Denominator population
1999	LBXSAL	5,445	826	15.170	mortality_eligible_before_complete_case_restriction
1999	LBXSCR	5,445	826	15.170	mortality_eligible_before_complete_case_restriction
1999	LBXGLU	5,445	3,168	58.182	mortality_eligible_before_complete_case_restriction
1999	LBXLYPCT	5,445	762	13.994	mortality_eligible_before_complete_case_restriction
1999	LBXMCV	5,445	732	13.444	mortality_eligible_before_complete_case_restriction
1999	LBXRDW	5,445	732	13.444	mortality_eligible_before_complete_case_restriction
1999	LBXSAPSI	5,445	826	15.170	mortality_eligible_before_complete_case_restriction
1999	LBXWBCSI	5,445	732	13.444	mortality_eligible_before_complete_case_restriction
2001	LBXSAL	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXSCR	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXGLU	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXLYPCT	5,987	717	11.976	mortality_eligible_before_complete_case_restriction
2001	LBXMCV	5,987	690	11.525	mortality_eligible_before_complete_case_restriction
2001	LBXRDW	5,987	690	11.525	mortality_eligible_before_complete_case_restriction
2001	LBXSAPSI	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXWBCSI	5,987	690	11.525	mortality_eligible_before_complete_case_restriction
2003	LBXSAL	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXSCR	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXGLU	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXLYPCT	5,610	591	10.535	mortality_eligible_before_complete_case_restriction
2003	LBXMCV	5,610	561	10.000	mortality_eligible_before_complete_case_restriction
2003	LBXRDW	5,610	561	10.000	mortality_eligible_before_complete_case_restriction
2003	LBXSAPSI	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXWBCSI	5,610	561	10.000	mortality_eligible_before_complete_case_restriction
2005	LBXSAL	5,561	586	10.538	mortality_eligible_before_complete_case_restriction
2005	LBXSCR	5,561	586	10.538	mortality_eligible_before_complete_case_restriction
2005	LBXGLU	5,561	3,137	56.411	mortality_eligible_before_complete_case_restriction
2005	LBXLYPCT	5,561	557	10.016	mortality_eligible_before_complete_case_restriction
2005	LBXMCV	5,561	535	9.621	mortality_eligible_before_complete_case_restriction
2005	LBXRDW	5,561	535	9.621	mortality_eligible_before_complete_case_restriction
2005	LBXSAPSI	5,561	586	10.538	mortality_eligible_before_complete_case_restriction
2005	LBXWBCSI	5,561	535	9.621	mortality_eligible_before_complete_case_restriction
2007	LBXSAL	6,219	655	10.532	mortality_eligible_before_complete_case_restriction
2007	LBXSCR	6,219	656	10.548	mortality_eligible_before_complete_case_restriction
2007	LBXGLU	6,219	3,468	55.765	mortality_eligible_before_complete_case_restriction
2007	LBXLYPCT	6,219	626	10.066	mortality_eligible_before_complete_case_restriction
2007	LBXMCV	6,219	612	9.841	mortality_eligible_before_complete_case_restriction
2007	LBXRDW	6,219	612	9.841	mortality_eligible_before_complete_case_restriction
2007	LBXSAPSI	6,219	657	10.564	mortality_eligible_before_complete_case_restriction
2007	LBXWBCSI	6,219	613	9.857	mortality_eligible_before_complete_case_restriction
2009	LBXSAL	6,510	561	8.618	mortality_eligible_before_complete_case_restriction
2009	LBXSCR	6,510	561	8.618	mortality_eligible_before_complete_case_restriction
2009	LBXGLU	6,510	3,581	55.008	mortality_eligible_before_complete_case_restriction
2009	LBXLYPCT	6,510	500	7.680	mortality_eligible_before_complete_case_restriction
2009	LBXMCV	6,510	491	7.542	mortality_eligible_before_complete_case_restriction
2009	LBXRDW	6,510	491	7.542	mortality_eligible_before_complete_case_restriction
2009	LBXSAPSI	6,510	561	8.618	mortality_eligible_before_complete_case_restriction
2009	LBXWBCSI	6,510	491	7.542	mortality_eligible_before_complete_case_restriction

Cycle start year	Biomarker	Mortality-eligible N	Missing N	Missing (%)	Denominator population
2011	LBXSAL	5,849	700	11.968	mortality_eligible_before_complete_case_restriction
2011	LBXSCR	5,849	700	11.968	mortality_eligible_before_complete_case_restriction
2011	LBXGLU	5,849	3,254	55.633	mortality_eligible_before_complete_case_restriction
2011	LBXLYPCT	5,849	545	9.318	mortality_eligible_before_complete_case_restriction
2011	LBXMCV	5,849	540	9.232	mortality_eligible_before_complete_case_restriction
2011	LBXRDW	5,849	540	9.232	mortality_eligible_before_complete_case_restriction
2011	LBXSAPSI	5,849	702	12.002	mortality_eligible_before_complete_case_restriction
2011	LBXWBCSI	5,849	540	9.232	mortality_eligible_before_complete_case_restriction
2013	LBXSAL	6,100	482	7.902	mortality_eligible_before_complete_case_restriction
2013	LBXSCR	6,100	482	7.902	mortality_eligible_before_complete_case_restriction
2013	LBXGLU	6,100	3,377	55.361	mortality_eligible_before_complete_case_restriction
2013	LBXLYPCT	6,100	430	7.049	mortality_eligible_before_complete_case_restriction
2013	LBXMCV	6,100	414	6.787	mortality_eligible_before_complete_case_restriction
2013	LBXRDW	6,100	414	6.787	mortality_eligible_before_complete_case_restriction
2013	LBXSAPSI	6,100	483	7.918	mortality_eligible_before_complete_case_restriction
2013	LBXWBCSI	6,100	414	6.787	mortality_eligible_before_complete_case_restriction
2015	LBXSAL	5,974	600	10.044	mortality_eligible_before_complete_case_restriction
2015	LBXSCR	5,974	601	10.060	mortality_eligible_before_complete_case_restriction
2015	LBXGLU	5,974	3,406	57.014	mortality_eligible_before_complete_case_restriction
2015	LBXLYPCT	5,974	564	9.441	mortality_eligible_before_complete_case_restriction
2015	LBXMCV	5,974	563	9.424	mortality_eligible_before_complete_case_restriction
2015	LBXRDW	5,974	563	9.424	mortality_eligible_before_complete_case_restriction
2015	LBXSAPSI	5,974	601	10.060	mortality_eligible_before_complete_case_restriction
2015	LBXWBCSI	5,974	563	9.424	mortality_eligible_before_complete_case_restriction

Note: Missingness was quantified within each cycle among mortality-eligible participants before complete-case restriction. The denominator is repeated for each biomarker within a cycle; biomarker-specific missing counts are not joint complete-case exclusions.

Provenance: tables/missingness_by_cycle.csv.

Supplementary Table S7. Mortality-eligible participants included in versus excluded from the benchmark cohort

Group	Population definition	N	Age mean	Age SD	Female (%)	Deaths N	Deaths (%)	Median follow-up (months)
benchmark_included	Complete for notebook 01 EXTENDED_16	19,829	47.009	19.050	50.643	2,936	14.807	124
excluded_before_benchmark	Missing one or more notebook 01 EXTENDED_16 fields	33,426	47.784	19.862	52.486	6,168	18.453	122

Note: Inclusion was defined by completeness for notebook 01's EXTENDED_16 fields. The two groups partition the 53,255 mortality-eligible participants: 19,829 included and 33,426 excluded.

Provenance: tables/cohort_included_excluded_characteristics.csv.

Supplementary Table S8. Common complete-case and training-derived-imputation sensitivity analyses

Analysis	Score	Cohort	N	Deaths	C-index	Age correlation	Missing-data rule	Status
common_complete_case	PhysioRisk	temporal_test	6,739	416	0.663	0.102	identical participants; all no-CRP comparator and PhysioRisk biomarkers observed	performed
common_complete_case	PhenoAge NCP (no-CRP proxy)	temporal_test	6,739	416	0.845	0.923	identical participants; all no-CRP comparator and PhysioRisk biomarkers observed	performed
common_training_derived_imputation	PhysioRisk	temporal_test	6,739	416	0.663	0.102	no-op in retained benchmark because all required biomarkers are observed	performed as degenerate no-missing sensitivity; extension to excluded participants not attempted
common_training_derived_imputation	PhenoAge NCP (no-CRP proxy)	temporal_test	6,739	416	0.845	0.923	no-op in retained benchmark because all required biomarkers are observed	performed as degenerate no-missing sensitivity; extension to excluded participants not attempted

Note: All 6,739 temporal-validation participants in the retained benchmark already had the biomarkers required for both scores. Common training-derived imputation was therefore a no-op and changed no observations. Extension to excluded participants was not attempted because doing so would require a separately specified missing-data strategy and would change the retained complete-case validation design.

Provenance: outputs/physiorisk_v2_validation/primary_validation_metrics_v2.csv.